

CANDIDATE
NAME

[illegible]CENTRE
NUMBER

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CANDIDATE
NUMBER

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9702/32

May/June 2024

2 hours

You must answer on the question paper.

You will need: The materials and apparatus listed in the confidential instructions

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You will be allowed to work with the apparatus for a maximum of 1 hour for each question.
- You should record all your observations in the spaces provided in the question paper as soon as these observations are made.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [].

For Examiner's Use	
1	
2	
Total	

This document has **12** pages. Any blank pages are indicated.

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You may not need to use all of the materials provided.

1 In this experiment, you will investigate an electrical circuit.

- (a) • Connect the circuit shown in Fig. 1.1.

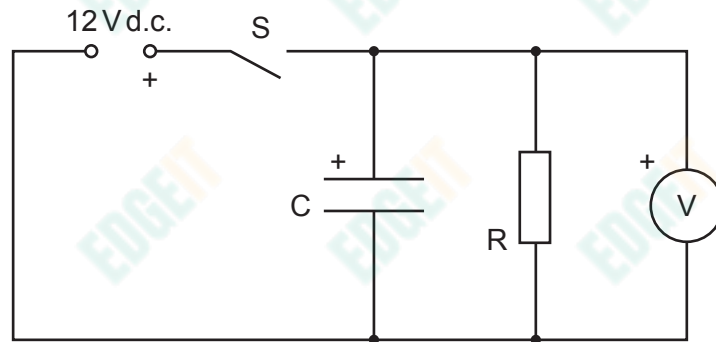


Fig. 1.1

- Ensure that the polarities of the power supply, component C and the voltmeter are as shown in Fig. 1.1.
- Close switch S for a short time and then open it.
- Watch the voltmeter reading as it reduces.

When the voltmeter reading passes a value V_S of 8.00 V, start the stop-watch.

When the voltmeter reading passes a value of 7.00 V, stop the stop-watch.

- Record the starting value V_S and the time T for the voltmeter reading to fall by 1.00 V.

$V_S =$

$T =$ [2]

- (b) Choose another starting value V_S . Close S for a short time and then open it. Measure the time T for the voltmeter reading to fall by 1.00 V from the starting value V_S .

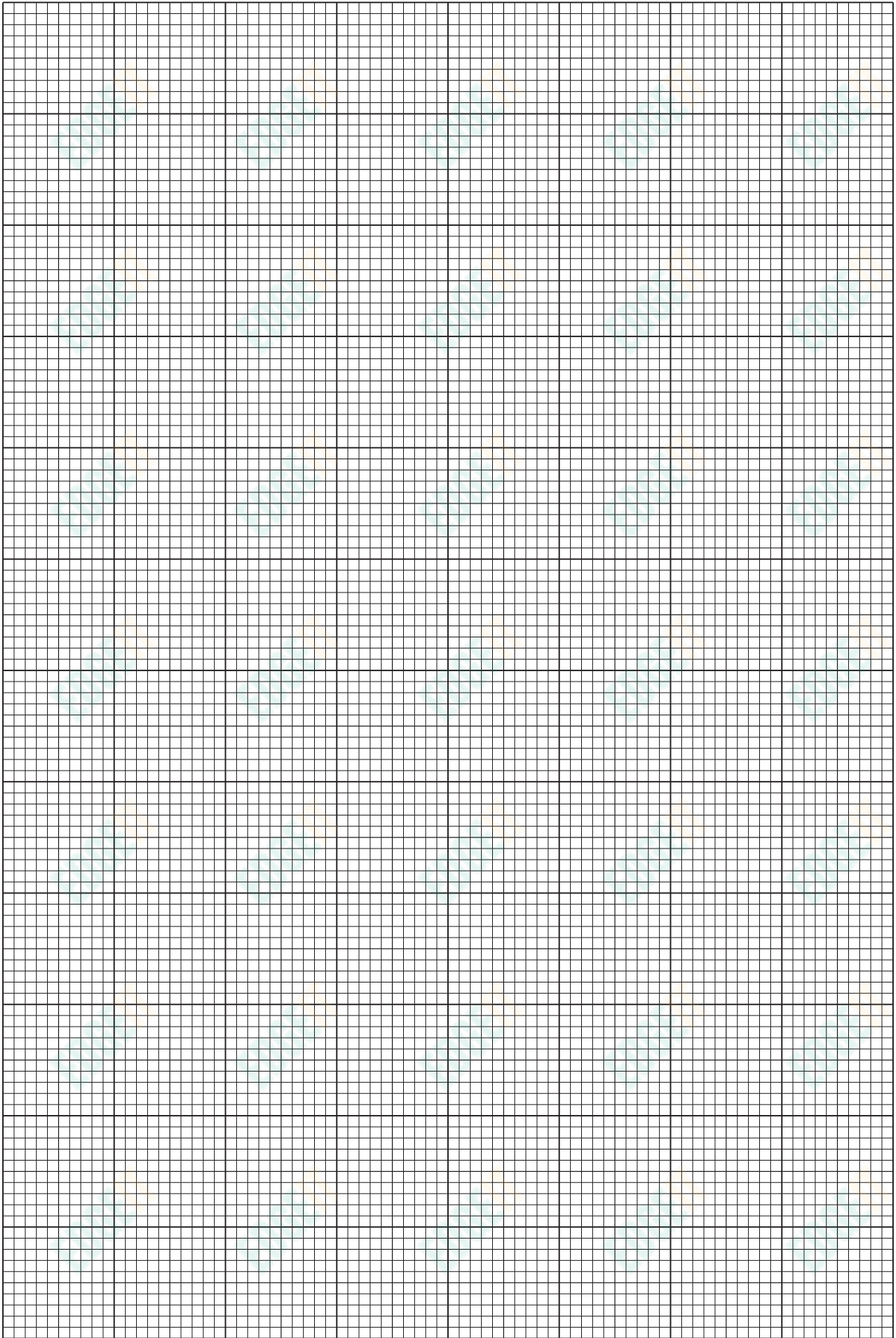
Repeat until you have six sets of values of V_S and T .

Record your results in a table. Include values of $\frac{1}{T}$ in your table.

- (c) (i) Plot a graph of $\frac{1}{T}$ on the y-axis against V_S on the x-axis. [10]
- (ii) Draw the straight line of best fit. [3]
- (iii) Determine the gradient and y-intercept of this line. [1]

gradient =

y-intercept = [2]



(d) It is suggested that the quantities V_S and T are related by the equation

$$\frac{1}{T} = aV_S + b$$

where a and b are constants.

Using your answers in (c)(iii), determine the values of a and b .
Give appropriate units.

$a =$

$b =$ [2]

[Total: 20]

You may not need to use all of the materials provided.

2 In this experiment, you will investigate the equilibrium of a wooden rod.

- (a) (i) • Assemble the apparatus as shown in Fig. 2.1.
- Adjust the apparatus so that the wooden rod is parallel to the bench and the spring is vertical.

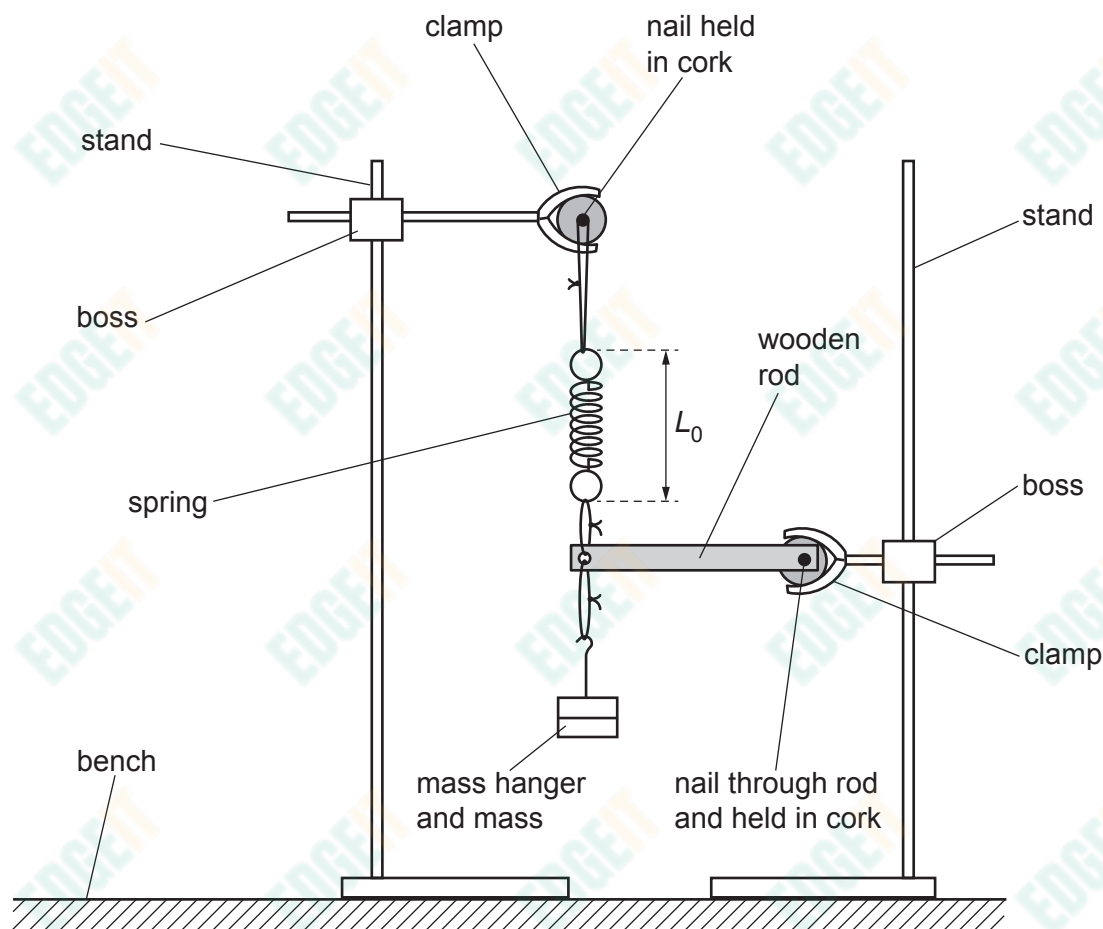


Fig. 2.1 (not to scale)

- The distance between the ends of the spring is L_0 , as shown in Fig. 2.1.
- Measure and record L_0 .

$L_0 = \dots\dots\dots$ m [1]

- (ii)
- Pull the mass hanger down a short distance and then release it. The mass hanger will oscillate.
 - Take measurements to find the period T of the oscillations.

$T =$ [2]

- (iii)
- Calculate the value of the spring constant k using

$$k = \frac{\alpha \pi^2}{T^2}$$

where $\alpha = 0.800 \text{ kg}$.

$k =$ Nm^{-1}

- Justify the number of significant figures that you have given for your value of k .

.....

[1]

- (b) (i) • Move the stand supporting the spring away from the other stand and add the plumb line, as shown in Fig. 2.2.

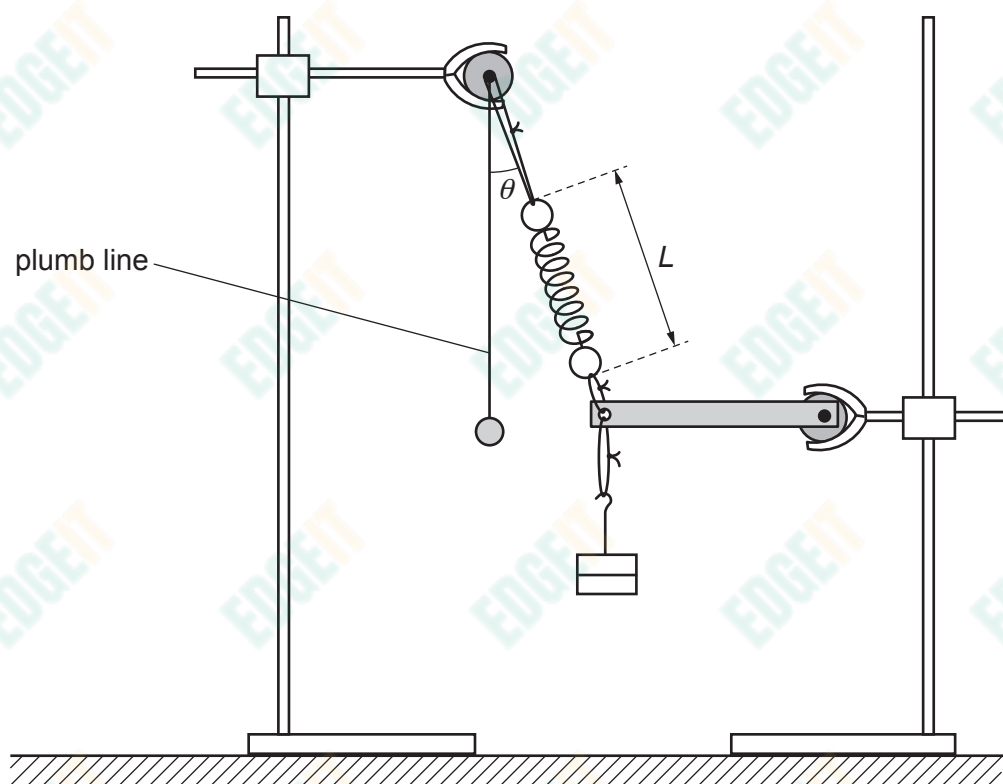


Fig. 2.2 (not to scale)

- Adjust the apparatus so that the angle θ between the spring and the vertical is approximately 20° and the wooden rod is parallel to the bench, as shown in Fig. 2.2.
- The new distance between the ends of the spring is L , as shown in Fig. 2.2.

Measure and record L .

$L = \dots\dots\dots$ m

- Measure and record θ .

$\theta = \dots\dots\dots^\circ$
[2]

- (ii) Estimate the percentage uncertainty in your value of θ . Show your working.

percentage uncertainty = $\dots\dots\dots\%$ [1]

(iii) Repeat (b)(i) using an angle θ of approximately 45° .

$L = \dots\dots\dots$ m

$\theta = \dots\dots\dots^\circ$
[3]

(c) It is suggested that the relationship between L and θ is

$$k(L - L_0) + B = \frac{D}{\cos \theta}$$

where $B = 2.0 \text{ N}$ and D is a constant.

Using your data, calculate two values of D .

first value of $D = \dots\dots\dots$

second value of $D = \dots\dots\dots$
[1]

(d) It is suggested that the percentage uncertainty in the values of D is 10%.

Using this uncertainty, explain whether your results support the relationship in (c).

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.....
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..... [1]

- (e) (i) Describe **four** sources of uncertainty or limitations of the procedure for this experiment.

For any uncertainties in measurement that you describe, you should state the quantity being measured and a reason for the uncertainty.

1

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2

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3

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4

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[4]

- (ii) Describe **four** improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

1

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2

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3

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4

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[4]

[Total: 20]

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